NEOSTIGMINE

Give with anticholinergic

Bradycardia risk

CHEMICAL NAME 3-{[(dimethylamino)carbonyl]oxy}-N,N,N-trimethylanilinium (as methylsulfate or bromide)

CLASS Reversible acetylcholinesterase inhibitor (parasympathomimetic)

TRADE NAMES Prostigmin · Bloxiverz · Neostigmine methylsulfate injection

THERAPEUTIC USES / INDICATIONS

- Reversal of non-depolarising neuromuscular block (atracurium, rocuronium) at end of surgery.
- Myasthenia gravis (diagnosis and treatment).
- Post-operative urinary retention and paralytic ileus (rarely).

ADMINISTRATION

- **IV** slowly over 1 min, always with glycopyrrolate or atropine to block muscarinic effects.
- **IM/SC** in myasthenia gravis.
- Give only when some spontaneous recovery is present (TOF ≥2 twitches).

DOSAGES

- **Reversal (adult):** 2.5 mg IV over 1 min (max 5 mg per dose; US label 0.03–0.07 mg/kg), given after or with glycopyrronium or atropine.
- **Children:** 0.05 mg/kg IV with anticholinergic.
- **Myasthenia (IM/SC):** 1–2.5 mg as needed.

PHARMACOKINETICS

- **Onset:** IV 1–3 min; peak 7–10 min. **Duration:** 1–2 h.
- Quaternary amine: does not cross blood–brain barrier.
- Metabolised by cholinesterase and liver; excreted in urine. **t½:** 50–90 min.

PHARMACODYNAMICS

- Inhibits **acetylcholinesterase** → ↑ ACh at neuromuscular junction and muscarinic sites.
- ACh competes with non-depolarising blockers → return of muscle strength.
- Muscarinic effects: bradycardia, secretions, bronchospasm, ↑ gut motility.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Bradycardia, salivation, sweating.
- Nausea, vomiting, abdominal cramps, diarrhoea.
- Miosis, lacrimation.

ADVERSE EFFECTS (SERIOUS)

- Bronchospasm.
- Arrhythmias, heart block, asystole (if given without anticholinergic).
- Cholinergic crisis in overdose: weakness, paralysis.

CONTRAINDICATIONS & CAUTIONS

- Mechanical bowel or urinary obstruction.
- Peritonitis.
- **Caution:** asthma, bradycardia, recent MI, epilepsy, Parkinson's disease.

KEY INTERACTIONS

- **Suxamethonium:** prolongs depolarising block.
- **β-blockers:** additive bradycardia.
- **Aminoglycosides:** antagonise its effect.

NURSING MANAGEMENT

- Confirm anticholinergic is drawn up and given together.
- Monitor HR, BP, ECG, SpO_2 and breathing adequacy.
- Assess reversal: sustained head lift, hand grip, tidal volume.
- Watch for secretions and bronchospasm; suction available.
- Myasthenia: record muscle strength before and after doses.

PATIENT EDUCATION

- Report increasing weakness, difficulty breathing or swallowing (myasthenia).
- Take myasthenia doses on time, before meals.

KEY POINTS

- Anticholinesterase; reverses non-depolarising block.
- Does not reverse suxamethonium.
- Always with glycopyrrolate or atropine.
- Bradycardia and secretions are main risks.

SOURCES CHECKED **[96]** Neostigmine Methylsulfate Injection BP 2.5mg in 1ml (UK SmPC) **[97]** Neostigmine, IV (US label) **[98]** Neostigmine, IV (US label, Dr.Reddy's Laboratories)

Full references p. 432–457